

STATSEI14 · PRACTICAL TUTORIAL

Deep Learning Meets **ETAS**

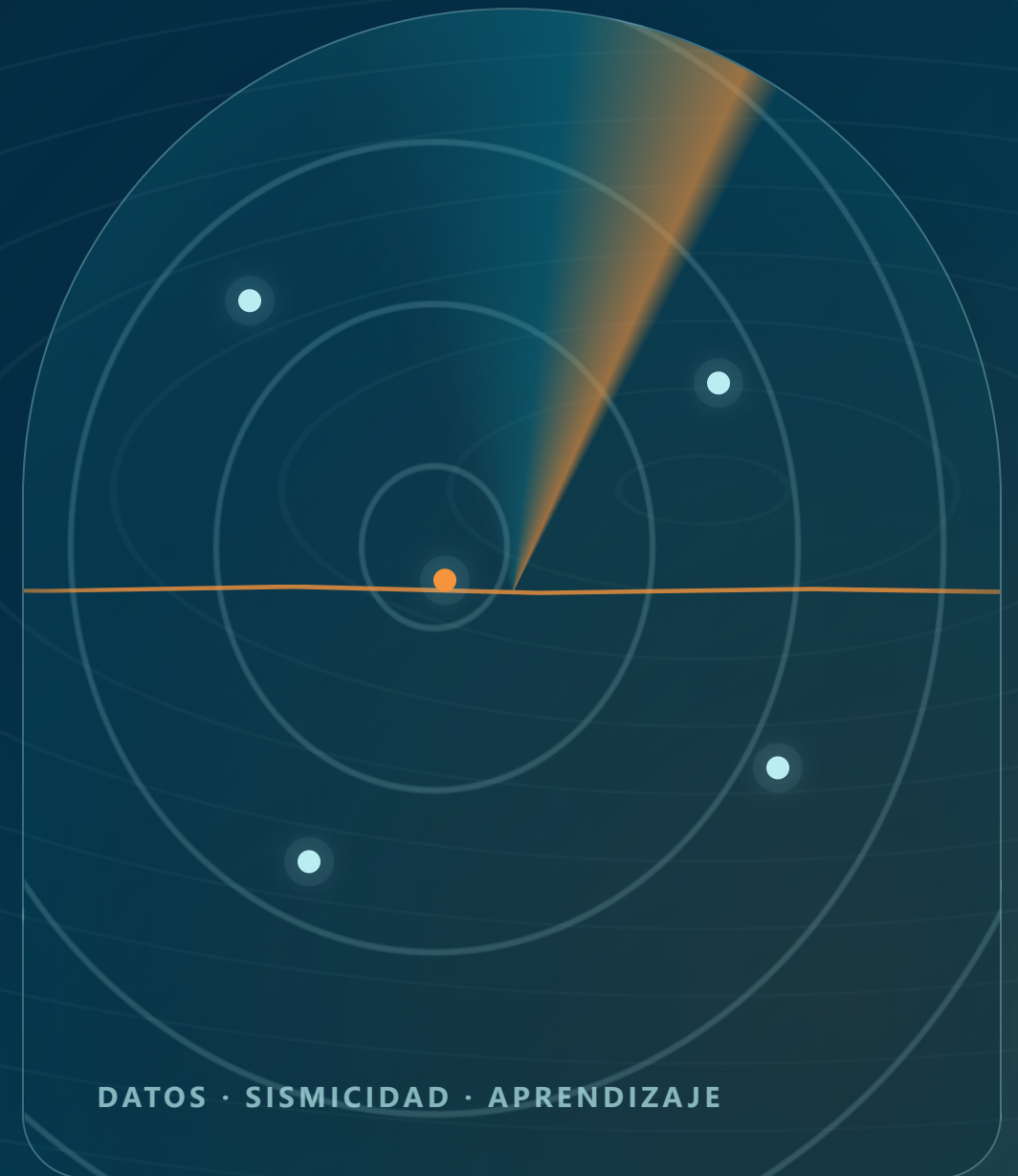
Practical Hybrid Workflows for Seismic Forecasting

Francisco Plaza-Vega

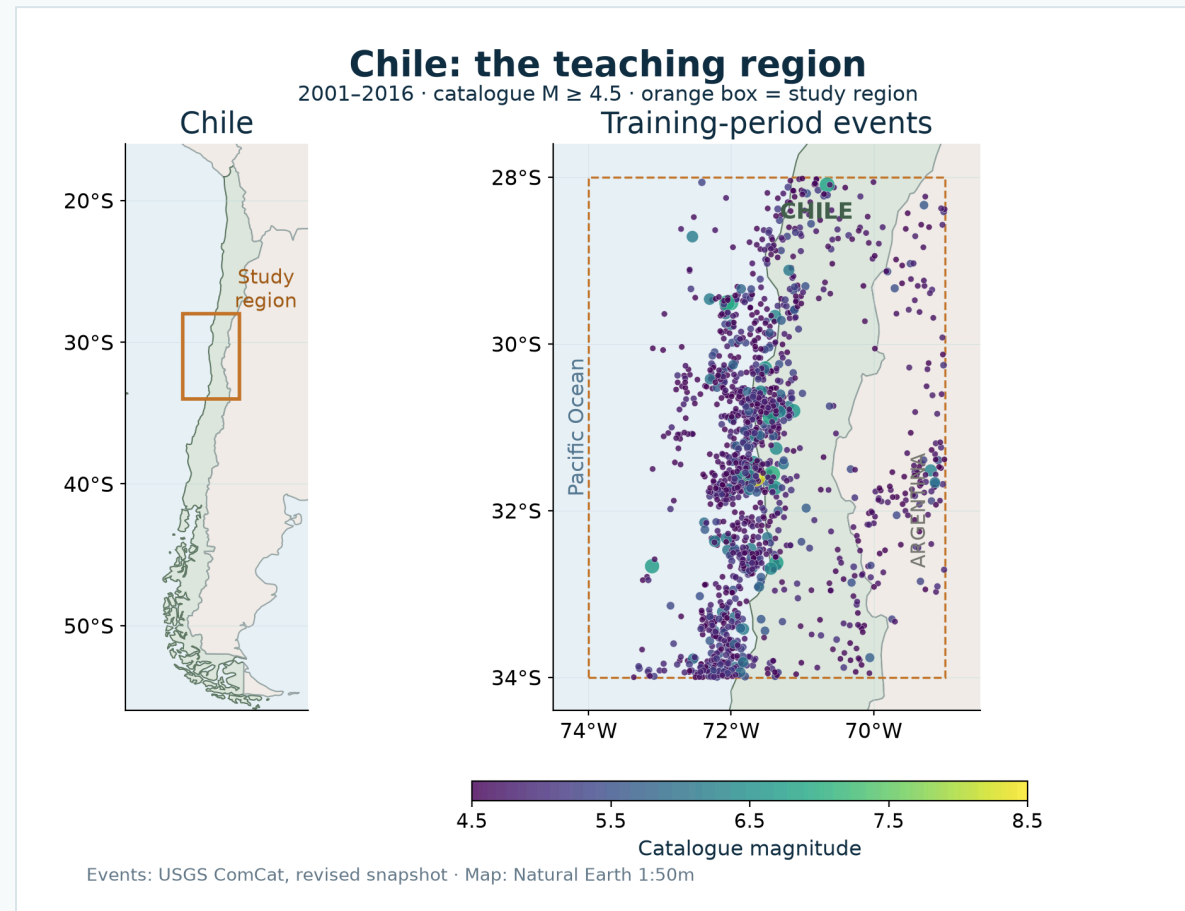
Universidad de Santiago de Chile

13 October 2026 · Chile

Learning about seismicity through representations, models and evidence.



What can we learn from seismicity?



Signals reveal recorded events.

Catalogues describe event histories.

Models turn histories into questions about what may follow.

What information do we represent, and what do we ask a model to learn?

Map: frozen USGS training catalogue and Natural Earth. Illustrates today's regional setting.

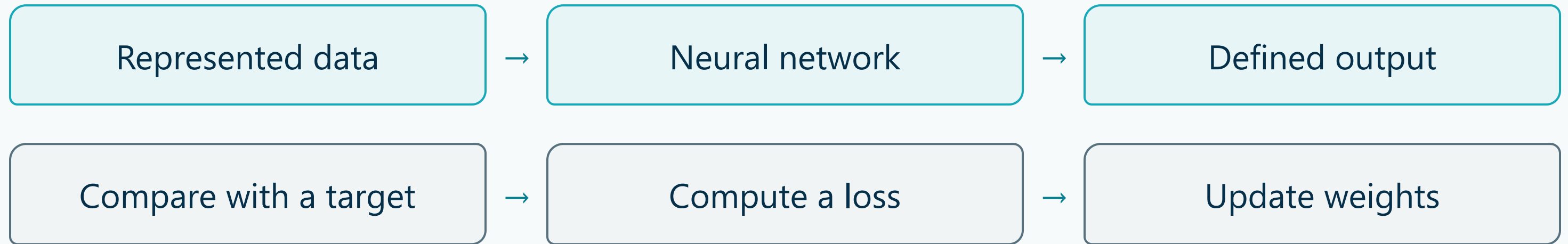
Deep learning addresses different seismic tasks

Example	Information supplied	Learning task
ConvNetQuake · 2018	Recorded waveforms	Detect events and classify their source region
RECAST · 2023	Catalogue event history	Model the timing of future occurrences
Transformers in Chile · 2025	Event features or sequences of grids	Classify events or forecast gridded activity

A useful model starts with a clearly defined question.

Perol et al. · Dascher-Cousineau et al. · Delgado et al. · Farfán et al.

A network learns a mapping from examples

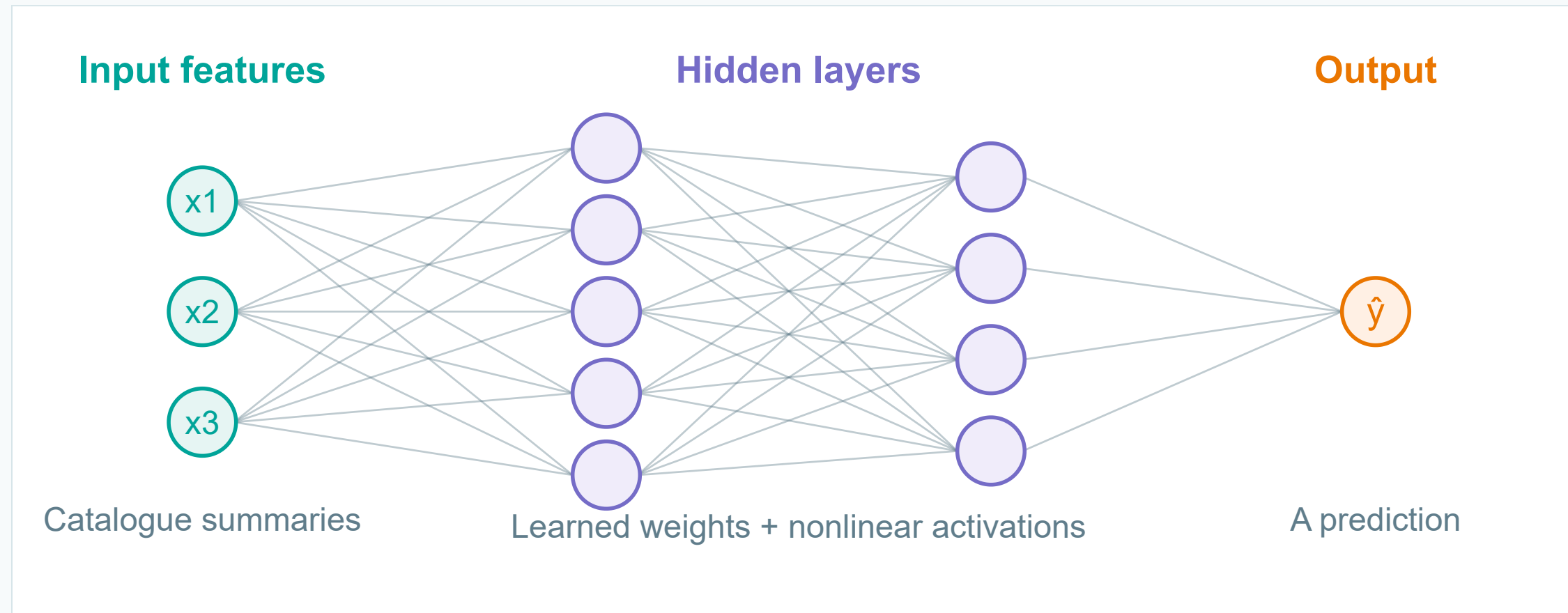


Architecture determines how information is combined.

Training adjusts weights to reduce the loss.

The output and the loss must match the scientific question.

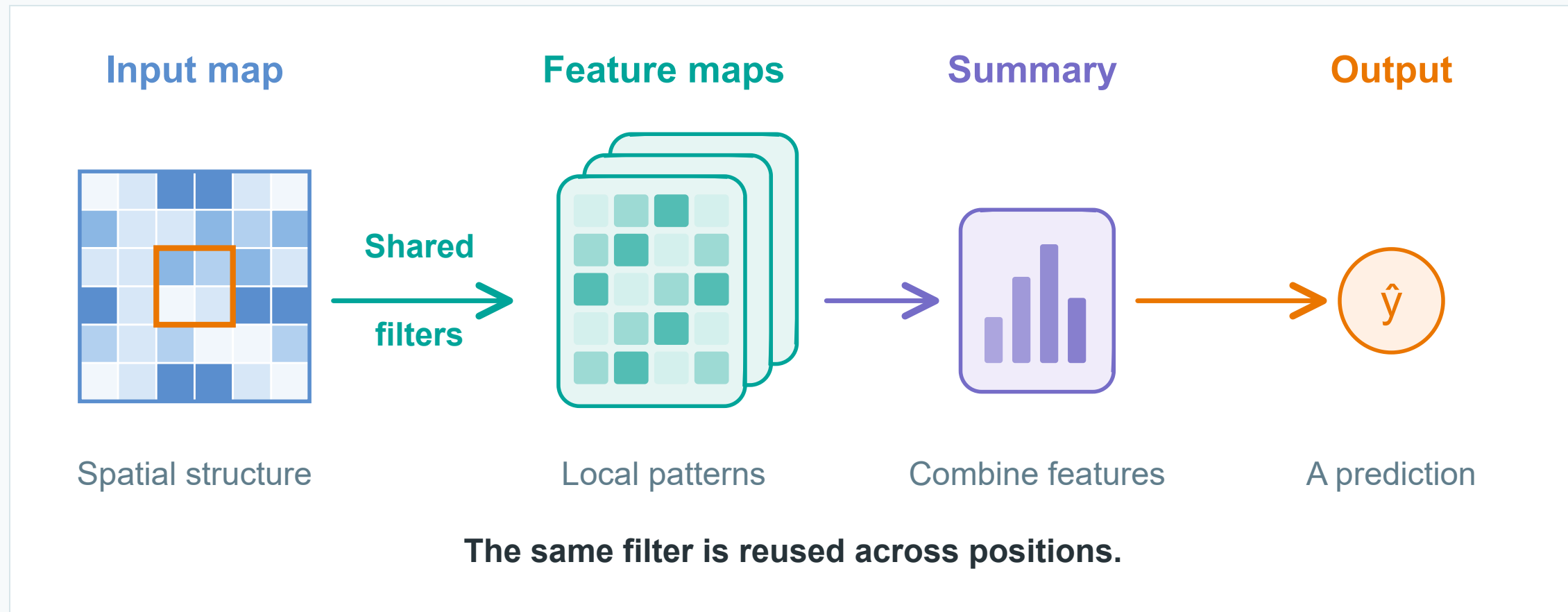
MLP: combine measured features



A multilayer perceptron learns nonlinear combinations of a feature vector.

Example input: recent count, magnitude summary and elapsed time. Adapted from the author's DL course, Unit 2.

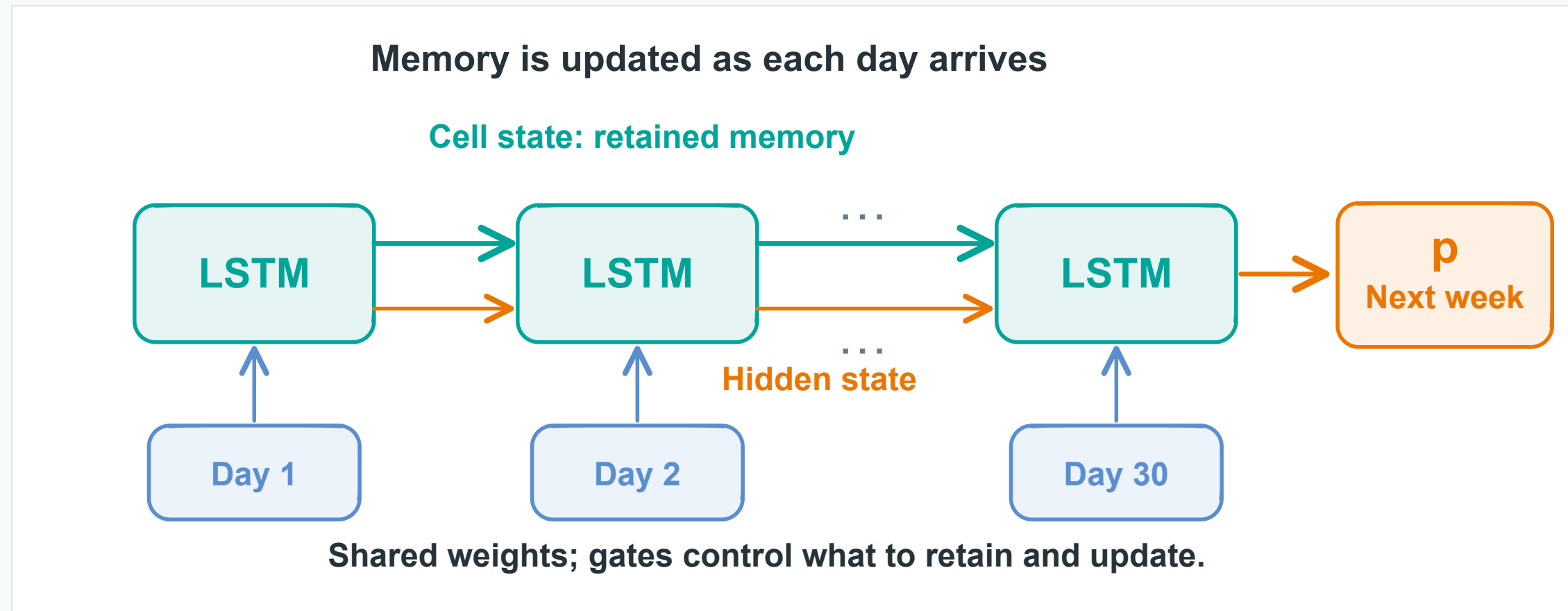
CNN: learn local patterns with shared filters



The same filter examines different neighborhoods.

Signals and spatial fields can both carry local structure. Adapted from the author's DL course, Unit 3.

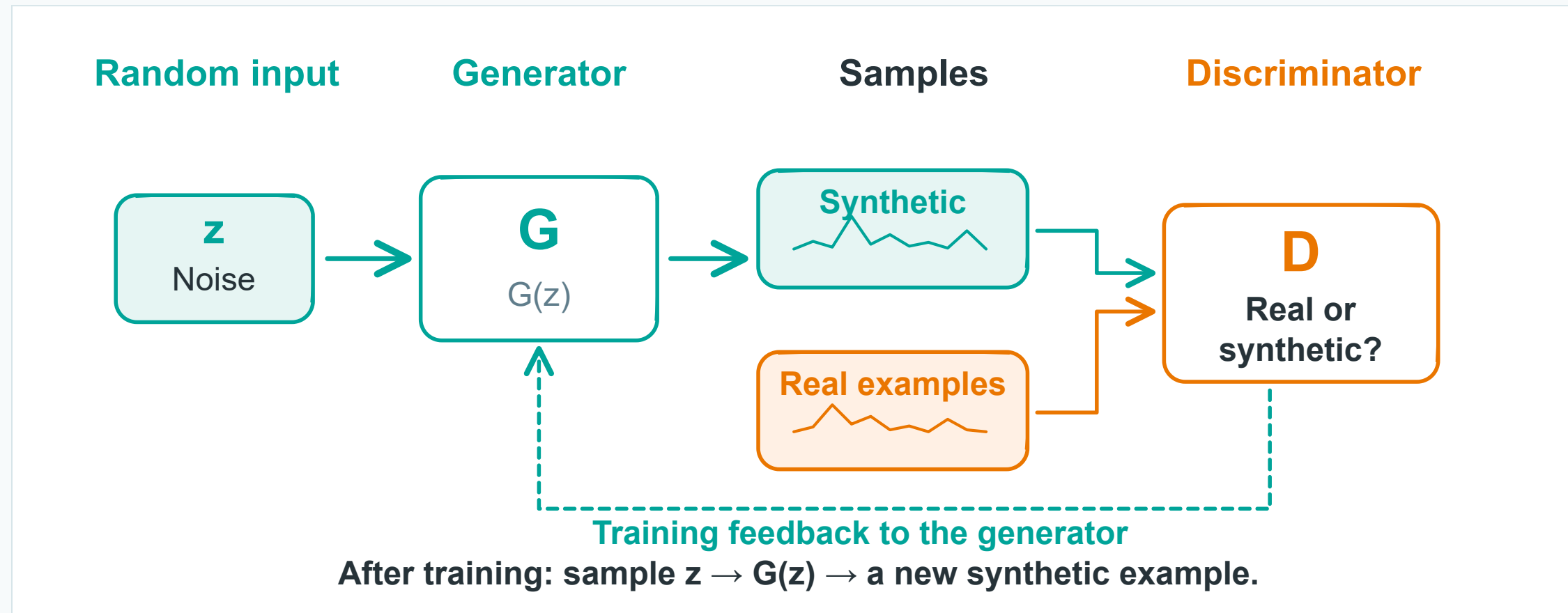
LSTM: carry information through a sequence



A learned state summarizes the history for a defined output.

Same learned update across time. Adapted from the author's DL course, Unit 4.

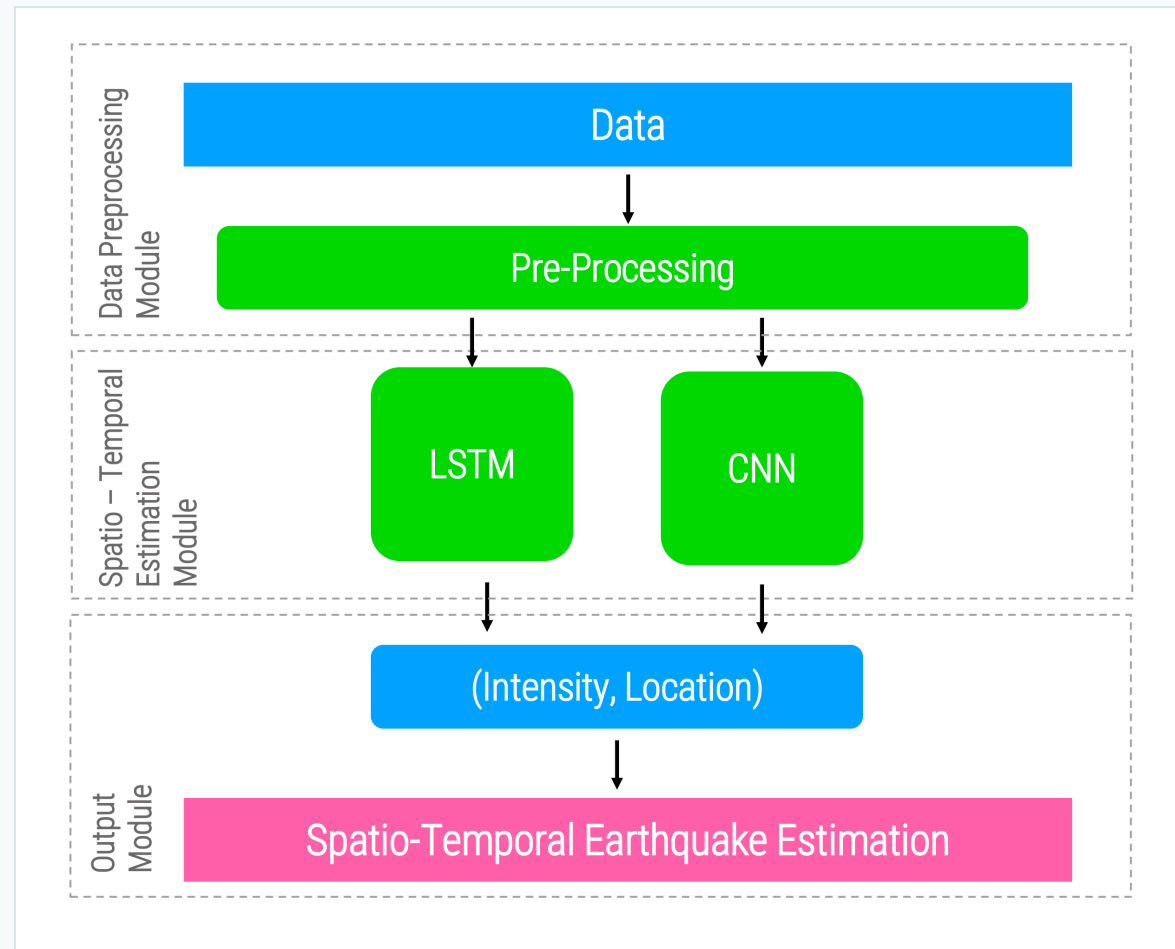
Generative learning: produce new samples



A generator learns to produce samples from a learned distribution.

GAN is one approach to generative learning. Adapted from the author's DL course, Unit 5.

Our first case: learning from an ETAS representation



Scientific question

Can neural networks learn the evolution of a statistical representation of seismic activity?

Two views

A daily time series and a sequence of spatial maps.

Nicolis, Plaza & Salas · Spatial Statistics 42 (2021), 100442 · CSN Chile, 2000–2017

ETAS supplies a statistical description of activity

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \text{contribution from event } i$$

Background + magnitude-dependent contributions that decay with time

Spatial ETAS also describes how contributions vary over space.

This conditional occurrence rate becomes a representation for learning.

Conceptual sketch. Process intensity is different from magnitude and shaking intensity.

The temporal task: tomorrow's maximum intensity



Input and target: summaries of the ETAS-derived intensity.

Learning question: how much of its evolution can a temporal network capture?

Specify what is being predicted before interpreting a fit statistic.

Published case, Nicolis, Plaza & Salas (2021). The target is a model-derived quantity.

The spatial task: the macrozone of the maximum

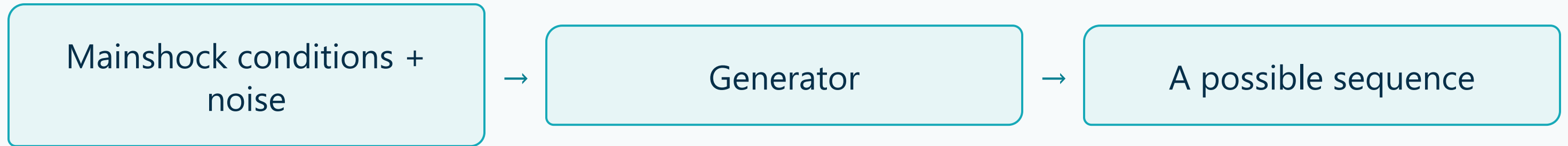


Which macrozone contains tomorrow's maximum estimated intensity?

Changing the representation and output opens a different learning task.

Published case, Nicolis, Plaza & Salas (2021). These probabilities concern six macrozones.

With Isidora: can we generate aftershock sequences?



Thesis question

Can conditional generative models reproduce the temporal pattern of aftershock sequences?

Model families explored

Conditional GAN · WGAN-GP · attention-based generator

A richer output requires us to decide what a sequence actually represents.

Isidora Jara Muñoz (2025), thesis · Subsequent manuscript by Plaza-Vega, Jara, Nicolis & Salinas (2026), under review

One sequence mixes absence and magnitude

Representation in the manuscript

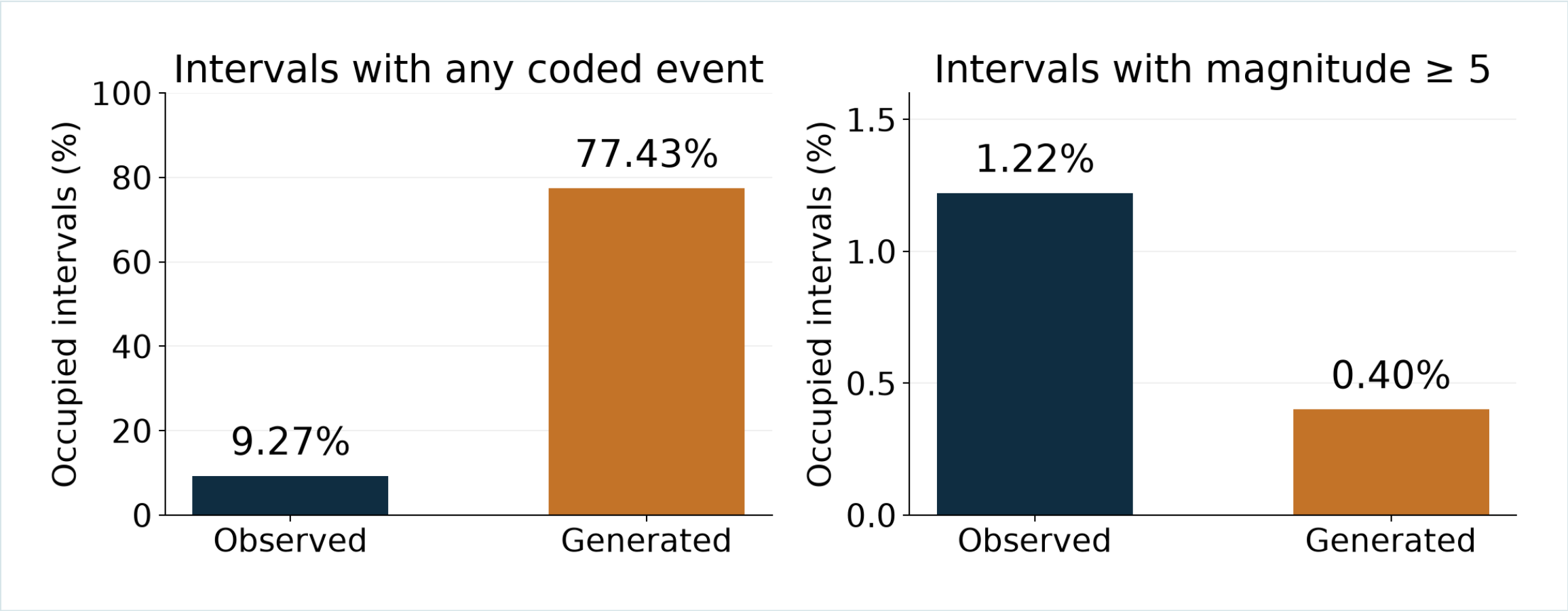
14 days → 224 intervals of 90 minutes → maximum magnitude per occupied interval



Schematic values. Gray: absence marker. Orange: maximum magnitude in an occupied interval.

“No recorded event” and “event magnitude” are different outcomes.

Generated sequences distort event occupancy



Assess which properties of the observed sequences are reproduced.

2026 manuscript under review · Occupied intervals, not total events · Source-case results

The article revisits what the model should learn

Empirical reformulation

Aggregate targets: occurrence, exceedance and count summaries.

The classification branch uses tree models.

A future neural formulation

Occurrence probability, followed by magnitude conditional on occurrence.

Evaluate both parts and their joint behavior.

Reformulating the target can matter as much as changing the architecture.

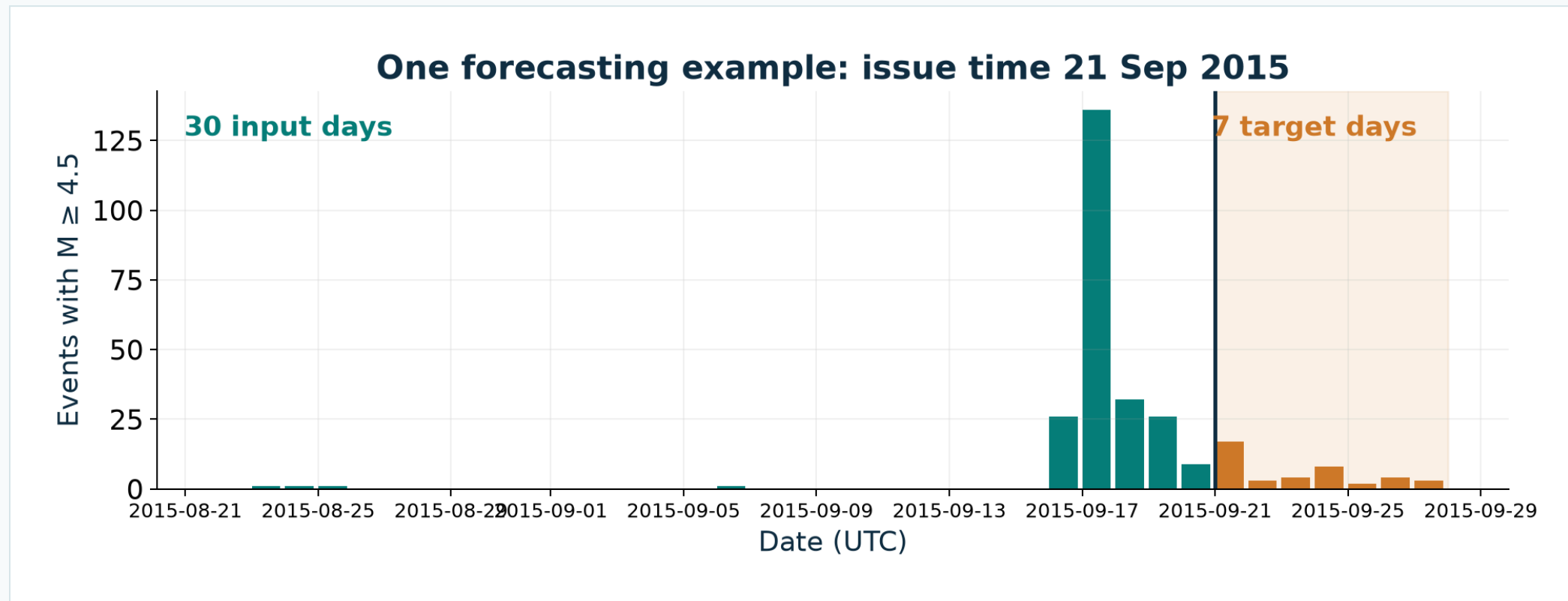
ETAS-informed generation remains future work. The two-part neural model is a proposal.

The practical puts the same decisions in your hands

Case	Representation	Output
Spatial Statistics	ETAS time series and maps	Intensity maximum and its macrozone
Isidora / manuscript	Mainshock conditions; binned sequences	Conditional sequences; aggregate targets
Today	Daily catalogue channels	Weekly occurrence probability

Represent the history → define the output → learn → check the evidence.

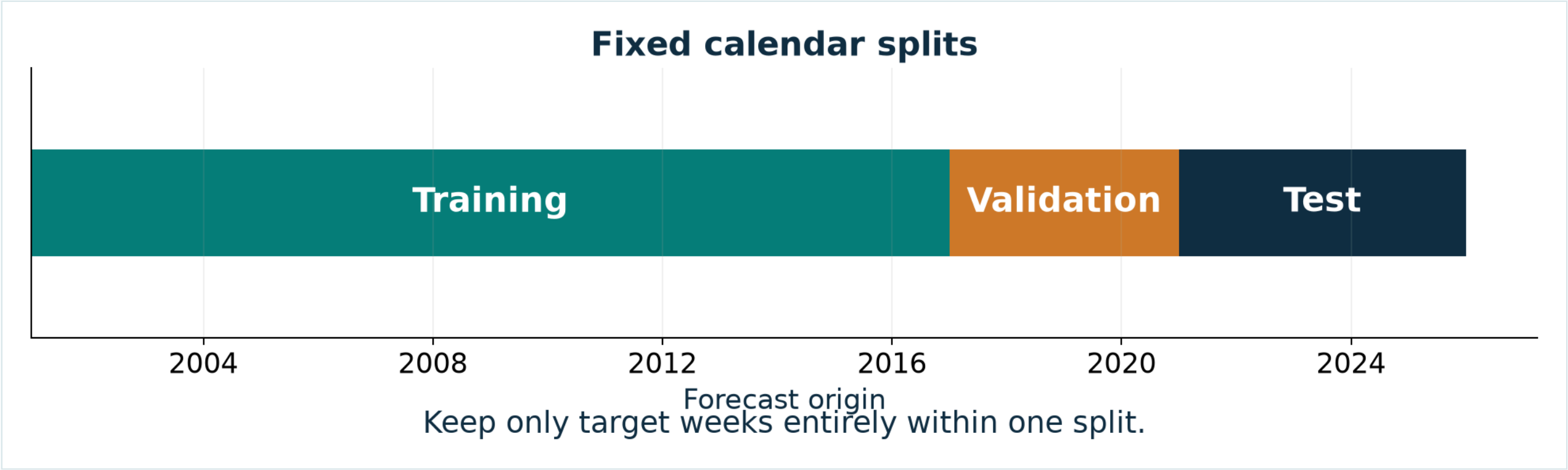
Today's target: at least one event next week



30 past days → probability of at least one catalogue $M \geq 5$ event in seven days.

Region: 28–34°S, 69–74°W · Input: catalogue $M \geq 4.5$ · Monday forecast origins

Evaluate using the temporal order of the problem



Compare probabilities with simple references using Brier score, log loss and calibration.

Train 2001–2016 · Validation 2017–2020 · Test 2021–2025 · Frozen, revised USGS catalogue

Experiment: does an informed representation help?

Catalogue channels → LSTM → occurrence probability

Catalogue channels + activity score → LSTM → occurrence probability

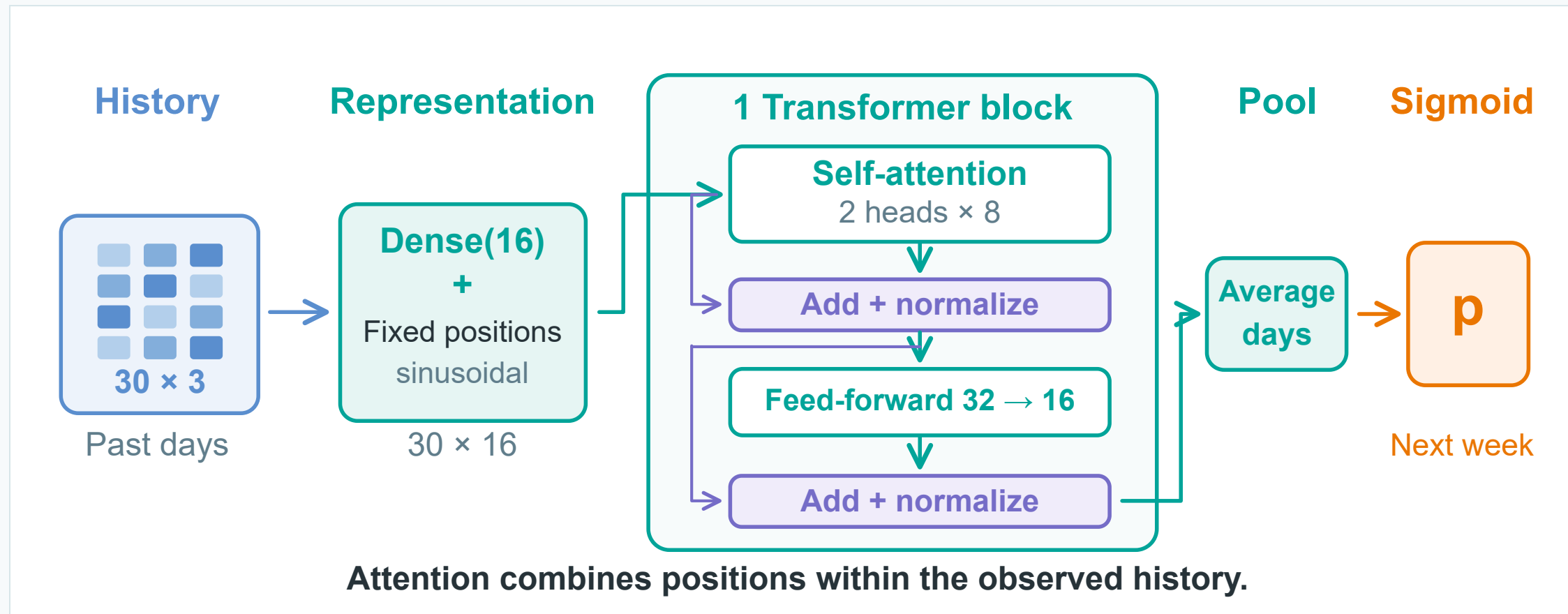
Before fitting: predict what the extra information might change.

After fitting: compare errors, calibration and simple baselines.

The score emphasizes larger, recent events with an illustrative decay.

Colab: [practical page](#) · Prepared results available · Transformer comparison follows the LSTM practice

A Transformer combines the same history differently



Same history and target. A different way to combine information across days.

Catalogue-only comparison · 2,305 trainable parameters versus 1,297 for LSTM · One fixed, small configuration

The Transformer did not improve this comparison

Model	Validation Brier	Test Brier	Test log loss
Training frequency	0.187690	0.165370	0.513204
Logistic regression	0.186611	0.164229	0.510317
LSTM	0.183065	0.165413	0.512942
Compact Transformer	0.188771	0.166367	0.515602

Lower is better. LSTM has the lowest validation Brier; logistic regression has the lowest test losses.

Changing the architecture gave us a useful comparison, without a gain in this pilot.

Pilot: 7 October 2026 · Neural scores are means over seeds 7, 17 and 27 · Catalogue-only comparison on the same split · Exploratory results on an already-inspected test period

What can this result tell us about the problem?

Observed in this pilot

The attention model did not improve the measured losses.

A simple baseline remained competitive.

Questions for the next experiment

Which representation and target preserve useful structure?

What information is available in the catalogue?

Would more data, capacity or a different training budget change the result?

Useful deep learning depends on how we represent the phenomenon and formulate the task.

These questions are hypotheses to investigate. The pilot does not identify the cause of its performance.

What did we learn, and what would we test next?

1. Which properties of the catalogue did our representation preserve?
2. What evidence supports changing an input or an architecture?
3. What would we evaluate next before making a broader claim?

Deep learning offers flexible tools. Their scientific value depends on the question and the evidence.

Homework: [guided reading](#) or [a controlled extension](#) · Build on today's LSTM and Transformer comparison

Appendix · a small Keras model

```
1 model = keras.Sequential([
2     keras.layers.Input(shape=(30, n_features)),
3     keras.layers.LSTM(16),
4     keras.layers.Dense(1, activation="sigmoid")
5 ])
6 model.compile(optimizer="adam", loss="binary_crossentropy")
7 history = model.fit(X_train, y_train,
8                     validation_data=(X_val, y_val),
9                     epochs=20, batch_size=64)
```

Sequential defines · compile configures · fit learns · predict applies

Appendix · quantities and scientific meaning

Quantity	Question answered
Process intensity	At what conditional rate do events occur?
Magnitude	How large is an event?
Shaking intensity	What motion or effects occur at a location?
Risk	What consequences may occur given exposure and vulnerability?

Today's output is a probability of a defined occurrence event.

Appendix · the exact activity score

$$r_t(s) = \sum_{t-L \leq t_i < s} 10^{\alpha(M_i - M_0)} \left(1 + \frac{s - t_i}{c}\right)^{-p}, \quad s \leq t.$$

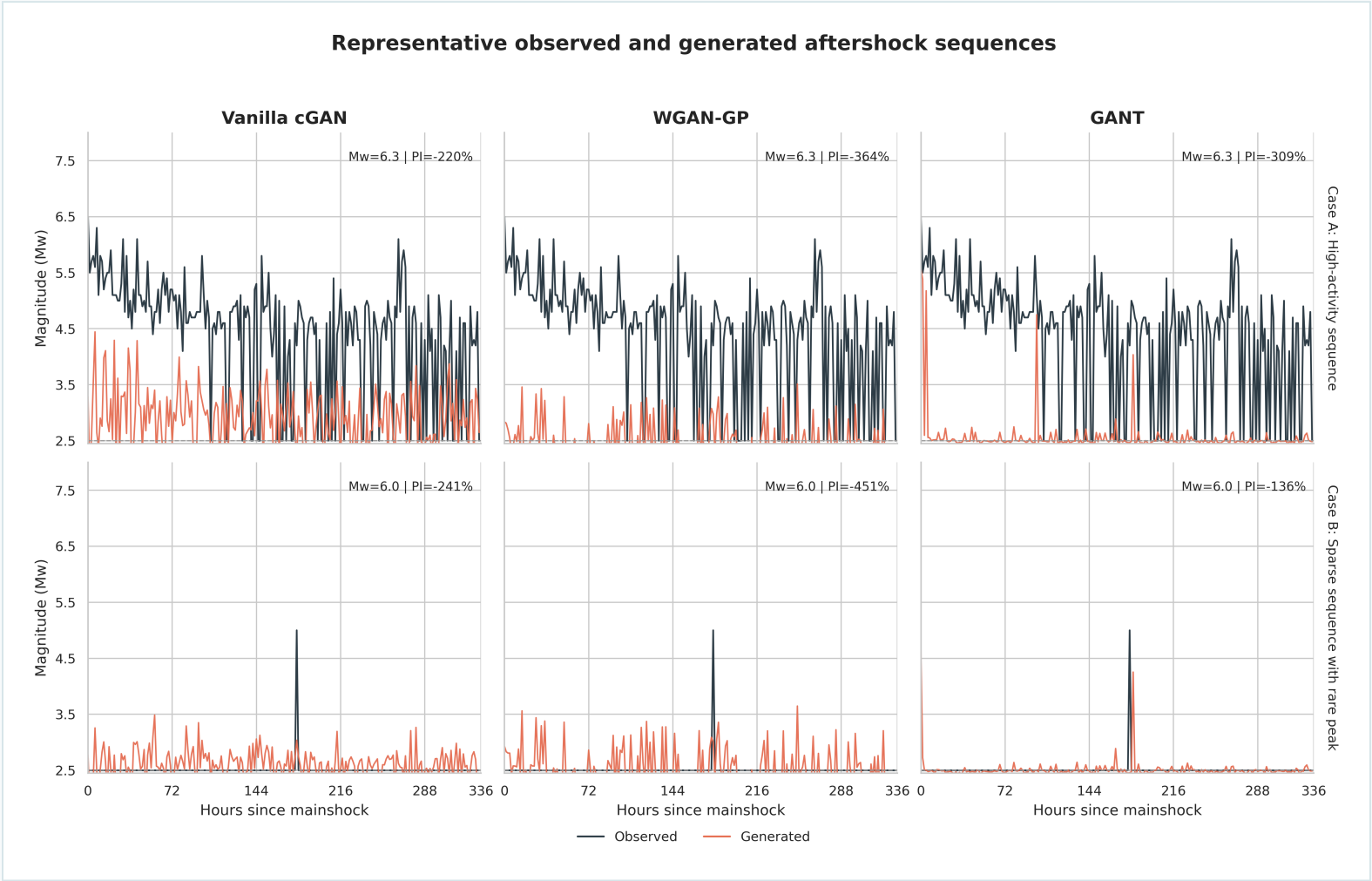
Illustrative: $\alpha=1$, $c=1$ day, $p=1.1$, $L=30$ days.

Shared history: the same available interval for both LSTMs.

Representation: $\log_{10}(\text{score})$ is an additional daily channel.

This index is neither an estimated ETAS rate nor a probability.

Appendix · original generative examples



2026 manuscript under review · Original PI annotations use a mean-reference denominator · Illustrative cases

Appendix · evidence and availability

Check	Meaning
Chronological, purged targets	Future target intervals do not cross partitions.
Training-only normalization	Later inputs do not set scaling parameters.
Frozen revised catalogue	Data bytes and provenance are reproducible.
Historical catalogue vintages	Not reconstructed.

Reproducibility and historical information availability are separate requirements.

Appendix · sources and further reading

- **Seismic applications:** [ConvNetQuake](#) · [RECAST](#).
- **Our published case:** [Nicolis, Plaza & Salas \(2021\)](#).
- **Generative case:** [Jara Muñoz \(2025\)](#), thesis; [Plaza-Vega, Jara, Nicolis & Salinas \(2026\)](#), manuscript under review.
- **Attention context and further reading:** [Delgado et al. \(2025\)](#) · [Farfán et al. \(2025\)](#).
- **Diagram style:** [author's DL course, Units 2–5](#) · Transformer schematic matches today's compact pilot.
- **Materials:** [practical](#) · [homework](#) · [full references](#).